REPORT SECTIONS ABOUT THE DATASET USED IN THIS STUDY

ABOUT THE DATASET

The dataset before training was divided into two parts: a **training set and a testing set**, where in *_70% where utilized to training set while 30% of the dataset where utilized to testing set_*. This approach ensures that the model is tested on unseen data to evaluate its performance.
Where in there are two data X and Y, X is the data variable and Y as the Label.
X ->> expressed x data into two arrays(training and testing)
We will trained our machine learning model with training data and
then we will evaluate the trained model within our testing data.
The trained learning model should have a capacity to predict what is within the testing data(accuracy).

Y->> expressed y data into two arrays(0) Non-diabetic and (1) Diabetic. Y train represent all labels for X train while Y test represent all labels within the test model.

XY parameters --> represent the old data of standardized data

STANDARDIZATION PROCESS OF THE DATA

Before the splitting of the dataset the data were first standardized before applying any classification models to the diabetes dataset, data standardization was performed to ensure that all features are on a similar scale. In this project, standardization was applied to the continuous variables, which include Glucose, BloodPressure, SkinThickness, Insulin, BMI, and Age. The standardization process involved transforming the features such that they have a mean of 0 and a standard deviation of 1. Once the features were standardized, the dataset was ready for splitting into training and test sets. We have a total of **768 original data and out of those 537 will be use within the training dataset and 231 will be done through the testing dataset**. The performance of the model was used with SVM, Kernel and Sigmoid model.

CLASSIFICATION RESULT

After standardizing the dataset, the Support Vector Machine (SVM) model was used for classification, with a sigmoid kernel. The SVM is a powerful classification algorithm that works by finding the optimal hyperplane that maximally separates the different classes in the feature space. The sigmoid kernel was chosen for its ability to handle non-linear data by transforming the input into a higher-dimensional space, where a linear decision boundary can be established.

Once the model was trained on the standardized dataset, it was evaluated on the test set, and performance metrics such as accuracy score, prediction score.It was found that **accuracy score of the training data is 68%. While the accuracy score of the testing data: 70%**. We can see the accuracy of testing data is higher than training data this means are dataset is not onto overfitting.Another scale method I used in classifying is the MinMax Scaler.In this section the features to a fixed range, typically [0, 1].With the same amount of percentage in training and testing data but the kernel used is linear model method with SVM. Min-Max scale garnered a high percentage both in training and testing compared with Standard Scale the training data of min_max is **78 while testing data got it a 75%**.With these result, linear kernel method has a more higher accuracy than kernel sigmoid. These mean that in data like these where class can be separated linear is more good to used than sigmoid.
It has a more simpler and efficent method having few hyperparemeter than sigmoid kernel. However Sigmoid is still used in these problem study as these problem has a two layer of network or have a little bit complex structures, therefore in a pivotal role where there is much more complext network, Sigmoid shines best than linear.The results highlighted how different scaling techniques impacted the performance of the classifier, providing insight into how well the models generalized to unseen data.But, overall they have the same prediction score.
Results
The model achieved the following performance on the training set:
Standard_Scale
accuracy Score: 68% prescion Score: 76%
Min_Max_Scale: accuracy Score: 78% precision Score: 76%
The model achieved the following performance on the testing set:
Standard_Scale
accuracy Score: 70%
Min_Max Scale
accuracy Score: 75%